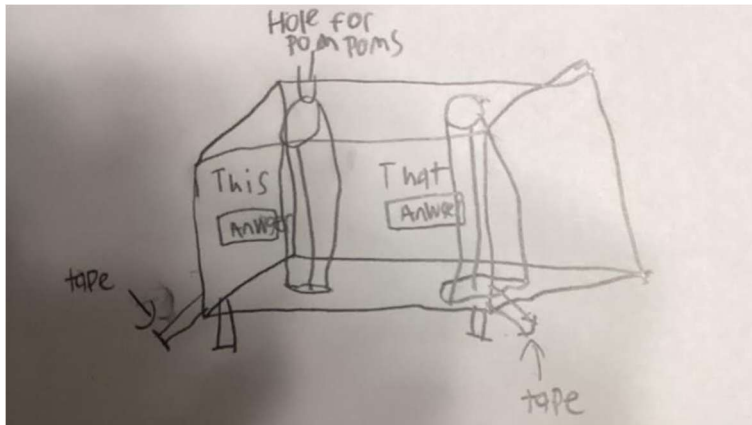


Element D: Design Concept Generation, Analysis, and Selection

Design ideas for the This or That station:

Design Idea #1

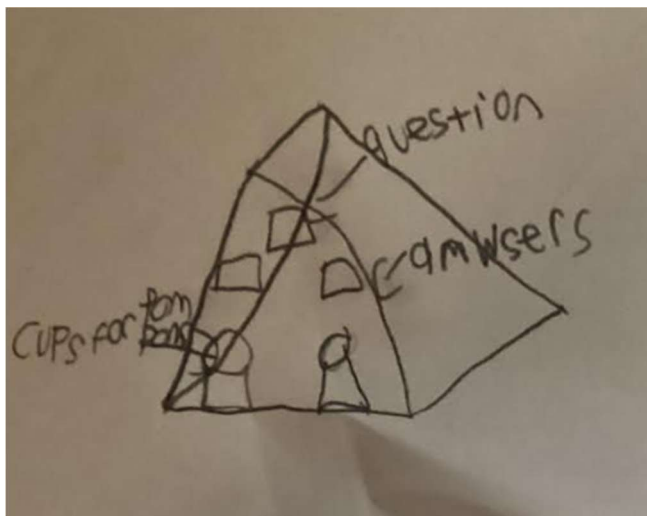
Date: 3/20/26



Description: For this design idea, there are 2 holes for each answer. There is a wall at the back with the 2 choices and 2 cylinders to place the pompoms (our choice for the object in which you put your choice). The front is empty to help the students see which choice is “winning.” Once the student wants to make their choice, they just place the pompom in either hole at the top.

Design Idea #2

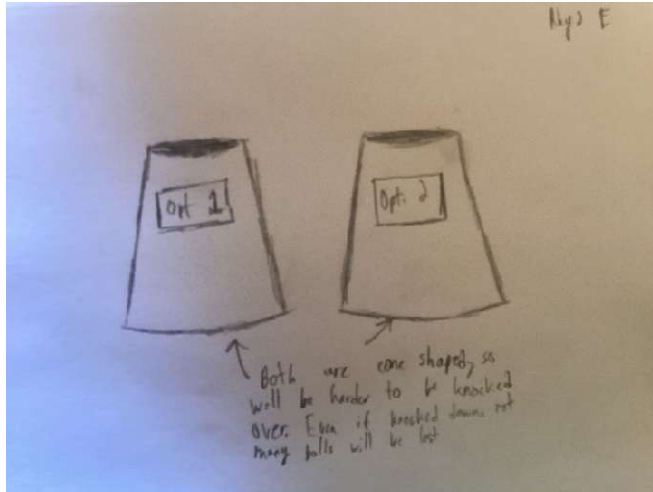
Date: 3/20/26



Description: For this design idea, there are two cups for each answer. Behind it are the questions with the answer's underneath it. The student can choose which one he/she likes better and cast their vote.

Design Idea #3

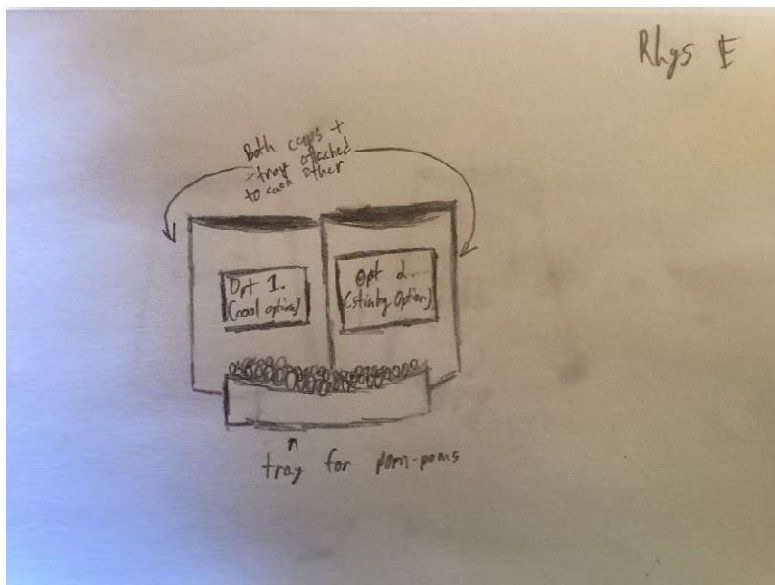
Date: 3/20/26



Description: For this design idea, there are still two cups for each answer. On each cup, there is an option, written on a piece of paper (which can be changed). The cups are cone-shaped, so it's harder to knock them over. In the case where they are knocked down, not many pompoms will be lost.

Design Idea #4

Date: 3/20/26

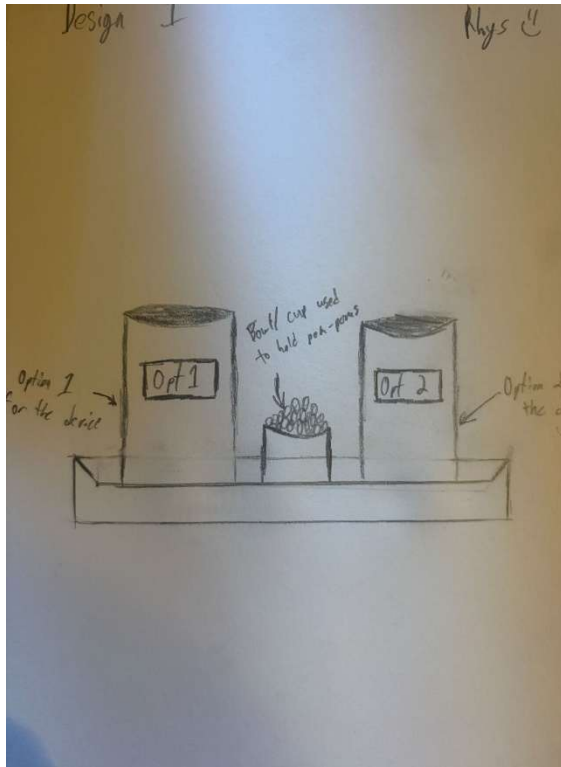


Description: For this design idea, there are two cups that are glued (or is 3d printer together) to a tray of pompoms. On each cup, there is an option written so students can choose which they like better. This way, the design is in one 1 can and thus can't break apart.

Design Idea # 5

Date: 3/20/26

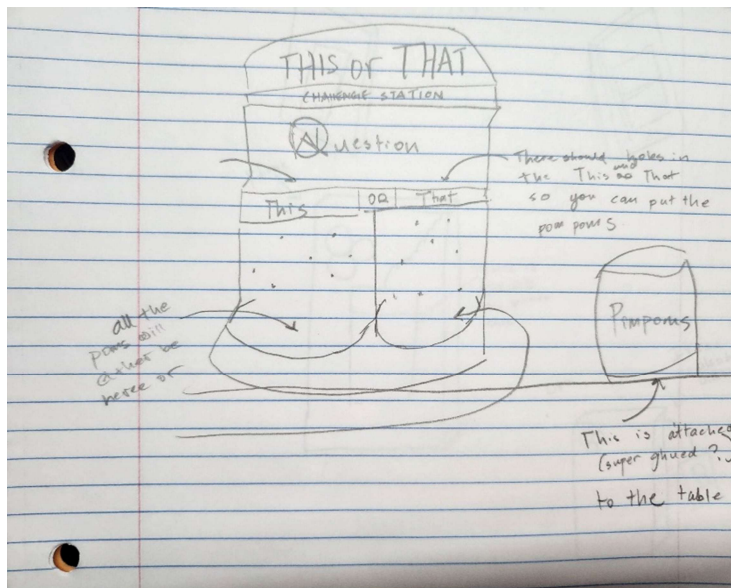
Description: This or that station design idea.



For this design idea, there are 2 cups with each option on it. In the middle, there is a smaller cup to put and take the pompoms. All of this is 3d printed on a tray (which makes the entire design more secure and in one piece).

Design Idea # 6

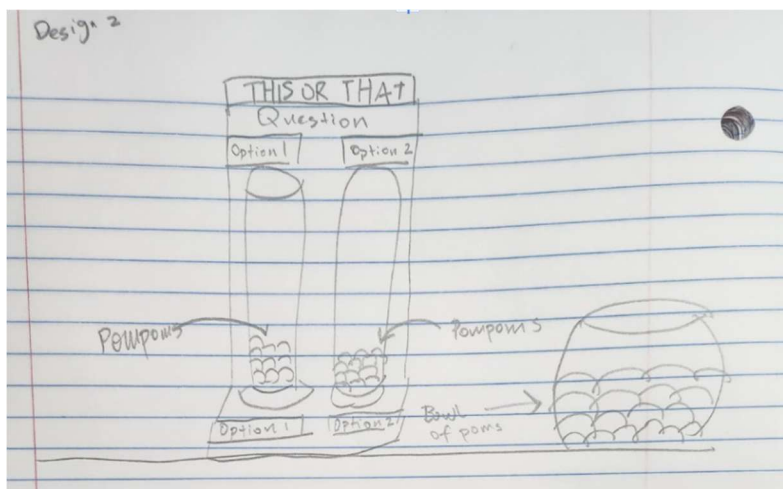
Date: 3/20/26



Description: For this design, it's supposed to be 3d printed. So, the entire thing is in one piece. At the top is the question, and underneath it is the options. There is a hole above each so students can place their votes. The pompoms will then fall down into two separate plates. Next to it is the jar of pompoms.

Design Idea # 7

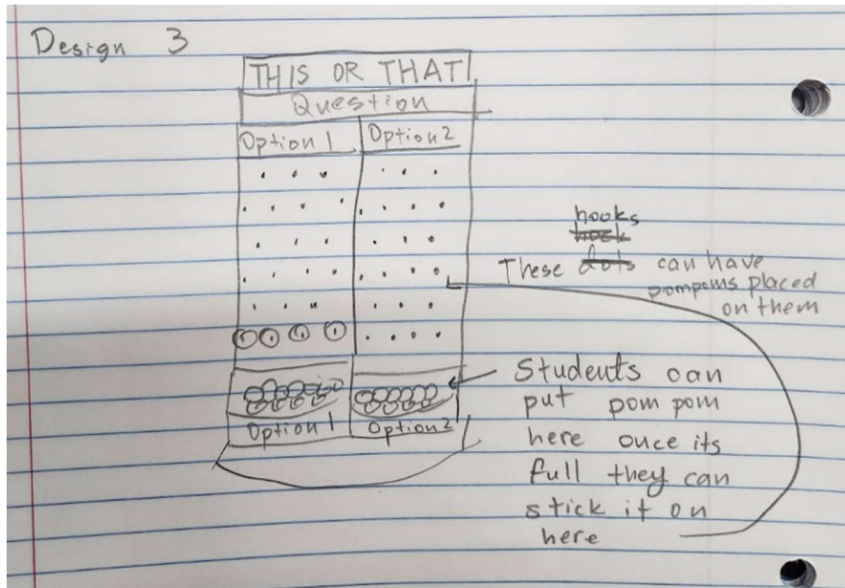
Date: 3/20/26



Description: For this design, there are two long jars to fill the pompoms. Above and below each is the option and students take the pompom from the bowl (right next to it) and fill whichever one they like.

Design Idea # 8

Date: 3/20/26



Description: For this design, there are 2 cups at each side (with the option below it). Once the cups are filled, the students can then place the pompoms on the hooks. There is a line in the middle of the two to help students differentiate between the two.

Design Idea Comparison and Selection (D1, D2)

Team Member 1's rating

	Weight	Design A	Design B	Design C	Design D	Design E	Design F
Must be 11 by 16 inches	0.3	1	1	0	1	1	1
Should be made in 1 piece	0.1	1	1	0	0	1	1
Must be made out of plastic	0.1	1	0	1	1	1	0
Cannot be small jars	0.2	1	0	1	1	0	1
Should be able to hold many pom-poms	0.2	1	1	1	1	1	1

Should be able to sit on a desk	0.1	-1	0	1	1	1	1
Score		0.8	0.6	0.6	0.9	0.8	0.9

Team Member 2's rating

	Weight	Design A	Design B	Design C	Design D	Design E	Design F
Must be 11 by 16 inches	0.1	1	1	0	1	1	1
Should be made in 1 piece	0.1	2	1	1	2	1	2
Must be made out of plastic	0.1	0	0	1	0	0	1
Cannot be small jars	0.2	1	1	1	1	1	0
Should be able to hold many pom-poms	0.2	1	1	1	0	1	-1
Should be able to sit on a desk	0.1	-1	0	-1	1	1	1
Score		0.8	0.8	0.5	0.8	0.9	0.5

Team Member 3's rating

	Weight	Design A	Design B	Design C	Design D	Design E	Design F
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Must be 11 by 16 inches	0.3	1	1	0	1	1	1
Should be made in 1 piece	0.1	2	1	0	0	1	1
Must be made out of plastic	0.1	1	0	1	1	1	1
Cannot be small jars	0.2	1	1	1	1	1	0
Should be able to hold many pom-poms	0.2	1	1	1	1	0	1
Should be able to sit on a desk	0.1	-1	-1	-1	1	-1	1
Score		0.9	0.7	0.4	0.9	0.6	0.8

Through our design matrix, my team chose design #5 as the final design.

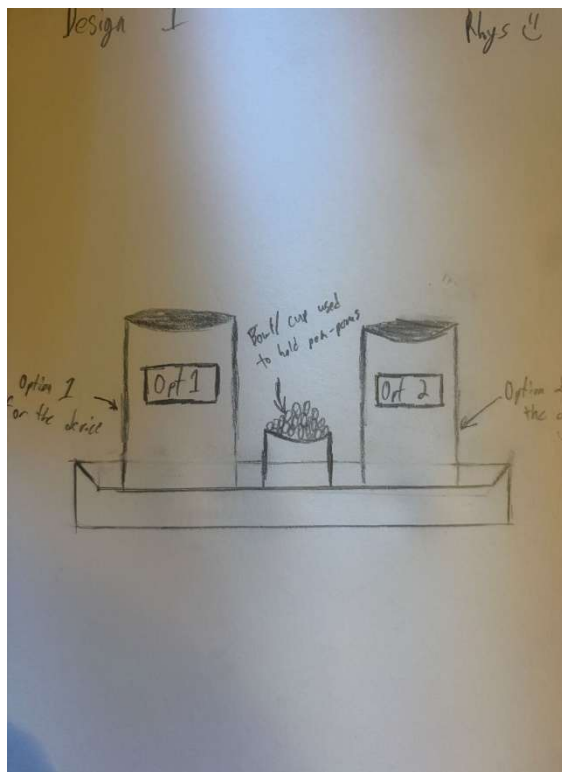
Design Blueprint (D4)

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Date: 3/20/26

Sketch:



Cad:

cad.onshape.com/documents/10a44b02d0dd4bf331b10502/w/2766da536eda2aa2cc25a3c2/e/33a3fbb23d007328d02358f4

Dimensions:

Base length: 9 inches

Base width: 4 inches

Base height: 1 inch

Large Cylinder diameter: 2.6 inches

Large Cylinder height: 5 inches

Center cylinder height: 2.7 inches

Materials:

3D printer filament (preferably see-through)

Pompoms

Manufacturing Methods:

Step 1: Export the CAD file as an STL file

Step 2: 3D print the file using a filament printer

Step 3:

Supply List and Costs:

Item	Cost each	# needed	Total
PLA Filament	~\$16 per 1 entire roll	>1 roll	~\$16

Total Cost: ~\$16

Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

Link: [Gantt Chart - Google Sheets](#)

Table1 										
Library Project Management	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14	Week 15	Week 16	Week 17	
Planning	  									
Find design										
Find materials										
Make the design diagram (cad)	  									
Write MyDeisgn D&H	  									
Prototype Making and Testing		  								
Make a prototype										
Test										
MyDesign G										
Preparation for Design-a-thon			  							
Miroboard presentation										
Feedback and Iterations										
MyDesign I & J				  						
Final Miroboard					  					
Miroboard presentation 2.0										
Reflection and portfolio										
MyDeisgn K&L						  				
Myself as an Engineer										
Enginneering for all video							  			
MyDeisgn Portfolio									  	
Student 1	Student 2	Student 3								